

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to
Advance Demand Flexibility
Through Electric Rates.

Rulemaking 22-07-005
(Filed July 14, 2022)

**REPLY COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON
ADMINISTRATIVE LAW JUDGE'S RULING ON TRACK B STAFF PROPOSAL TO
EXPAND EXISTING PILOTS**

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”) and the email ruling extending comment deadline issued by Administrative Law Judge (“ALJ”) Stephanie Wang on September 13, 2023, the Vehicle-Grid Integration Council (“VGIC”) hereby submits these reply comments on *Administrative Law Judges Ruling on Track B Staff Proposal to Expand Existing Pilots* (“Ruling”), issued by the Commission on August 15, 2023.

I. INTRODUCTION.

VGIC reiterates its appreciation for Energy Division (“ED”) staff for proposing that the Commission direct Investor-Owned Utilities (“IOUs”) to expand certain existing dynamic rate pilots, as described in the *Staff Proposal on Existing Dynamic Rate Pilot Expansion* (“Staff Proposal”) issued by ED staff on August 15, 2023.¹ VGIC’s reply to opening comments, summarized below, aims to further enhance the proposed expansions by recommending measures that enable electric vehicles (“EVs”) to provide demand flexibility and grid exports.

¹ CPUC R.22-07-005. *Administrative Law Judges Ruling on Track B Staff Proposal to Expand Existing Pilots*. August 15, 2023.

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M517/K407/517407755.PDF>

- VGIC supports the Joint Parties’ proposal to expand a CalFUSE-style rate to bundled and unbundled customers, which is more appropriately sized to the need for demand flexibility than Energy Division Staff expansion proposal #2.
 - VGIC agrees with the Joint Parties proposal elements intended to expand customer access through expanded rate eligibility and included programs.
 - VGIC strongly recommends the Commission adopt Joint Parties’ proposal to include non-NEM export compensation equal to the generation and distribution import price.
 - VGIC agrees with Joint Parties that allowing some flexibility on the exact design of the CalFUSE distribution and subscription components is warranted. However, the need for flexibility should be balanced with a reasonable degree of certainty for participating customers, aggregators, and installers.
 - VGIC recommends that the Distribution Investment Deferral Framework (“DIDF”) Partnership Pilot be deemed an included program, in addition to the supply-side and load-modifying programs proposed by Joint Parties.
 - VGIC respectfully requests that Energy Division staff and PG&E work to rename the “subscription” component in either the dynamic rate pilot expansion proposal or PG&E’s Business Electric Vehicles (“BEV”) rate, as each rate uses the term differently.
- To yield any near-term participation from vehicle-to-grid (“V2G”) resources, the smart inverter exemption in place for certain V2G resources must apply to SCE’s and PG&E’s Dynamic Rate Pilots.

II. VGIC SUPPORTS THE JOINT PARTIES’ PROPOSAL TO EXPAND A CALFUSE-STYLE RATE TO BUNDLED AND UNBUNDLED CUSTOMERS, WHICH IS MORE APPROPRIATELY SIZED TO THE NEED FOR DEMAND FLEXIBILITY THAN ENERGY DIVISION STAFF EXPANSION PROPOSAL #2.

In opening comments, GridX, Inc., Polaris Energy Services, Gridtractor Inc., and PG&E (together “Joint Parties”) presented and described in detail a more ambitious version of staff’s proposal #2 to expand Valley Clean Energy’s (VCE)/PG&E’s Agricultural Flexible Irrigation Technology (“AgFIT”) Pilot to additional customer groups. In opening comments, the Joint Parties highlight the specific components, terms, and scope necessary to successfully implement ED Staff proposal #2 at a more appropriate scale and with a few key pilot design modifications. Joint Parties

refer to the expansion of AgFIT to bundled and unbundled residential, commercial, and industrial customers across PG&E's service territory as the "PG&E CalFUSE Pilot" (Pilot #2)."²

VGIC commends the Joint Parties for proposing the PG&E CalFUSE Pilot #2, which VGIC agrees would "meet and further serve the objectives of the ED Staff's pilot expansion proposal by making a CalFUSE Pilot offering available to more customers".³ VGIC acknowledges the scale of the Joint Parties' proposal is an order of magnitude greater than ED staff's proposal. However, given the urgent need for demand flexibility to support California's evolving grid and ambitious policy goals, VGIC views the significantly expanded scope as crucial to supporting the grid. Considering the California Energy Commission's Load Management Standard, which requires utilities to offer optional CalFUSE-style rates by 2027, VGIC believes it is prudent to scale up CalFUSE-style pilots today to yield meaningful lessons for the full-scale implementation in just a few years. VGIC posits this would reduce risk for ratepayers, as full-scale, optional dynamic rates, if not first carefully tested, can negatively impact customers and, more importantly, may lead to unintended cost shifts.

A. VGIC agrees with the Joint Parties proposal elements intended to expand customer access through expanded rate eligibility and included programs.

VGIC generally supports expanding PG&E CalFUSE Pilot #2 to additional residential customers (i.e., EV2-A) and small commercial customers (i.e., B-6 and B-10). VGIC also strongly supports the expanded list of eligible programs that customers may dual-enroll in under PG&E's CalFUSE Pilot #2, which extends beyond staff's list to include the Emergency Load Reduction Program ("ELRP") and Demand Side Grid Support ("DSGS") program. Joint Parties accurately

² Joint Parties opening comments, Pg. 2.

³ *Ibid.*

state that “excluding customers on these programs is likely to seriously limit enrollment in and learnings from proposed Pilot #1 and #2.”⁴ VGIC recommends the Commission adopt the expanded customer eligibility terms in PG&E CalFUSE Pilot #2.

B. VGIC strongly recommends the Commission adopt Joint Parties’ proposal to include non-NEM export compensation equal to the generation and distribution import price.

Critically, PG&E’s CalFUSE Pilot #2 seeks to incorporate non-NEM export compensation, which marks a major enhancement relative to ED staff’s expansion proposal #2 and would provide a significant step forward toward ED staff’s CalFUSE vision. VGIC reiterates that bidirectional charging solutions represent a major source of latent energy storage capacity waiting to be unlocked to support grid reliability, renewable energy integration, community resiliency, and a more cost-effective utilization of the grid. VGIC, therefore, strongly supports the Joint Parties’ recommendation to add a non-NEM export compensation component that applies the same generation and distribution import price to exports as it does to imports.

C. VGIC agrees with Joint Parties that allowing some flexibility on the exact design of the CalFUSE distribution and subscription components is warranted. However, the need for flexibility should be balanced with a reasonable degree of certainty for participating customers, aggregators, and installers.

In opening comments, the Joint Parties propose using the same dynamic rate elements as AgFIT, which includes a dynamic distribution rate component and a subscription element that are generally aligned with the CalFUSE framework outlined in the ED White Paper issued on June 22, 2022.⁵ However, in opening comments, the Joint Parties state that each component is “subject

⁴ Joint Parties opening comments, Pg. 19.

⁵ CPUC Energy Division White Paper and Staff Proposal. *Advanced Strategies for Advanced Demand Flexibility Management and Customer DER Compensation*. June 22, 2022.

to change during the pilot based on additional analysis.”⁶ VGIC understands that these components may require enhancements throughout the pilot period and agrees that this type of mid-stream modification is appropriate for rates in the pilot stage of development. For example, the subscription element could utilize a baseline determined by the prior year’s hourly load as envisioned in PG&E’s Vehicle-Grid Integration (“VGI”) Pilots, a monthly average baseline from the prior year as utilized in PG&E’s current AgFIT Pilot, or could incorporate flexibility for the customer to elect their own baseline as recommended in opening comments of the California Large Energy Consumers Association (“CLECA”).⁷ VGIC believes the second and third examples may be simpler to implement, provide more opportunity to optimize EV charging and discharging, and, in the case of CLECA’s approach, provide a high degree of optionality to customers. Regarding dynamic distribution components, as VGIC notes in opening comments, non-rate programs may provide value to utilities, customers, and ratepayers alike, and may offer a helpful alternative to circuit-specific pricing.

With this in mind, VGIC agrees that PG&E should be permitted a certain degree of flexibility to modify and enhance the subscription, distribution, and other components of the pilot where necessary. However, this flexibility must be balanced with a moderate degree of certainty for participating customers, aggregators, and installers. These customers, aggregators, and installers may make significant upfront investments in technologies to enable pilot participation based on expected rate component design. VGIC recommends the Commission should consider how best to balance these needs for implementation flexibility and customer certainty by ensuring

<https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/demand-response/demand-response-workshops/advanced-der---demand-flexibility-management/ed-white-paper---advanced-strategies-for-demand-flexibility-management.pdf>

⁶ Joint Parties opening comments, Pg. 17.

⁷ CLECA opening comments, Pg. 6.

stakeholders, including participating or prospective customers, aggregators, and installers, are brought in early and often to discuss changes to the pilot methodology.

D. VGIC recommends that the Distribution Investment Deferral Framework (“DIDF”) Partnership Pilot be deemed an included program, in addition to the supply-side and load-modifying programs proposed by Joint Parties.

VGIC reiterates its strong support for including ELRP and DSGS in the “included programs” list where dual participation with PG&E CalFUSE Pilot #2 is permitted. In opening comments, the California Efficiency and Demand Management Council (“CEDMC”) highlighted that “[t]his pilot period is an ideal opportunity to develop and test performance measurement methodologies that would enable dual participation models.” CEDMC then notes that given the limited scale of the Energy Division staff’s proposed expansion, the possibility of overpaying dual-participation customers would be less significant, especially compared to the opportunities to support the grid.⁸ While PG&E CalFUSE Pilot #2 would expand the eligibility to additional rate classes and include programs than the ED Staff proposal, VGIC still agrees with the CEDMC’s statement and believes that PG&E CalFUSE Pilot #2 proposal is the appropriate, controlled venue to pilot dual participation at this time.

The Joint Parties list most supply-side and load-modifying demand response programs as eligible for dual participation with the PG&E CalFUSE Pilot #2. However, VGIC notes that the Distribution Investment Deferral Framework (“DIDF”) Partnership Pilot is not listed.⁹ VGIC urges the Commission to include the DIDF Partnership Pilot as an additional “included program” for customers, as the Partnership Pilot is an important pathway for identifying and deploying non-wires solutions to traditional distribution system upgrades.

⁸ California Efficiency and Demand Management Council opening comments, Pg. 6.

⁹ Joint Parties opening comments, Pg. 19.

E. VGIC respectfully requests that Energy Division staff and PG&E work to rename the “subscription” component in either the dynamic rate pilot expansion proposal or PG&E’s BEV rate, as each rate uses the term differently.

The proposals from ED staff and the PG&E CalFUSE Pilot #2 proposed by the Joint Parties, both use a subscription component option based on customer-specific load shapes, which is in line with Element 5 of the CalFUSE framework outlined in the ED White Paper, issued on June 22, 2022.¹⁰ However, VGIC is concerned that the use of the term “subscription” could confuse some participating customers given that PG&E’s BEV rate uses the same “subscription” term to define a different component. This can lead to confusion for customers, particularly commercial fleet customers who may have sites subject to two different types of subscription.

According to the CalFUSE framework outlined in the ED White Paper, “a subscription load shape is billed at the “legacy” rate and any deviation from the subscription load shape is billed/credited to the customer based on the CalFUSE price signal”¹¹ and in PG&E’s Electric Schedule BEV, the term “subscription” is defined as follows:

“SUBSCRIPTION: The kW subscription component of the BEV replaces a kW demand charge. The BEV rate does not include demand or customer charges and instead uses a monthly subscription for customer’s kW demand. The subscription comes in blocks of 10 kW for the BEV-1 rate option and in blocks of 50 kW for the BEV-2 rate option. A customer determines their estimated maximum kW demand and then selects a subscription level to suit their need.”¹²

¹⁰ CPUC Energy Division White Paper and Staff Proposal. *Advanced Strategies for Advanced Demand Flexibility Management and Customer DER Compensation*. June 22, 2022.

<https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/demand-response/demand-response-workshops/advanced-der---demand-flexibility-management/ed-white-paper---advanced-strategies-for-demand-flexibility-management.pdf>

¹¹ CPUC Energy Division White Paper and Staff Proposal. *Advanced Strategies for Advanced Demand Flexibility Management and Customer DER Compensation*. June 22, 2022. Pg. 66.

<https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/demand-response/demand-response-workshops/advanced-der---demand-flexibility-management/ed-white-paper---advanced-strategies-for-demand-flexibility-management.pdf>

¹² PG&E Electric Schedule BEV – Business Electric Vehicles. Sheet 5.

https://www.pge.com/tariffs/assets/pdf/tariffbook/ELEC_SCHEDULE_BEV.pdf

With this in mind, VGIC recommends stakeholders begin using an alternative term for either CalFUSE or the BEV rate to avoid confusion for participating customers.

III. TO YIELD ANY NEAR-TERM PARTICIPATION FROM VEHICLE-TO-GRID (“V2G”) RESOURCES, THE SMART INVERTER EXEMPTION IN PLACE FOR CERTAIN V2G RESOURCES MUST APPLY TO SCE’S AND PG&E’S DYNAMIC RATE PILOTS.

The Joint Parties’ opening comments note that the PG&E CalFUSE Pilot #2 proposal closely aligns with the ED Staff proposal to expand the AgFIT pilot, which VGIC generally agrees with. However, one area where the PG&E CalFUSE Pilot #2 proposal comes short of aligning with the ED Staff proposal is the lack of a UL 1741 SA/SB exemption for V2G DC electric vehicle supply equipment (“EVSE”). The ED Staff Proposal proposes applying the UL 1741 SA/SB certification exemption as adopted by Decision (“D.”) 21-12-015 to the proposed pilots in recognition of the nascent market for V2G EVSE.¹³ As demonstrated by the CEC’s V2G Equipment List, there are currently two SA-certified V2G DC EVSE and zero SB-certified V2G DC EVSE.¹⁴ Simply put, if UL 1741 SA and SB are required for V2G DC EVSE to participate in dynamic rates and non-NEM export participation, the utilities will see *no* customer participation for some time.¹⁵

Without this exemption, many bidirectional EVSE will remain latent grid support assets when they are most needed and available to provide grid flexibility services. Moreover, for the

¹³ CPUC R.22-07-005. *Administrative Law Judges Ruling on Track B staff Proposal to Expand Existing Pilots – Attachment A: Staff Proposal on Existing Dynamic Rate Pilot Expansion*. August 15, 2023. Pg. 12-13.

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M517/K408/517408172.PDF>

¹⁴ CEC. *V2G Equipment List*. Accessed October 9, 2023. <https://v2gel.energy.ca.gov/Home/ProcessView>

¹⁵ VGIC estimates SB-certified V2G EVSE may be available in 18-24 months, based on our discussions with bidirectional charging equipment manufacturers that are seeking UL 1741 SB certification.

nascent V2G services and bidirectional EVSE market to grow and experience market transformation, customers and project developers need access to more programs that offer export-based compensation that improve the business case for V2G projects. The proposed expansion of IOU dynamic rate programs that offer compensation for V2G exports is commendable, as previously only ELRP offered compensation for V2G exports. However, if these programs are not accessible to the vast majority of VGI aggregators using UL 1741 and UL 1741-SA certified EVSE, but not UL 1741-SB certified EVSE, these programs will not support the business case for V2G technology to scale. SCE and PG&E's dynamic rate pilots represent **significant near-term opportunities to demonstrate the potential for VGI to provide grid services at scale and be compensated for doing so. We therefore urge the Commission to make note of the current state of the bidirectional charger market and consider ways to support customers with UL 1741 and UL1741-SA certified V2G bidirectional chargers.**

To that end, VGIC reiterates its support for this exemption and urges the Commission to apply it to near-term SCE and PG&E's dynamic rate pilots. Moreover, VGIC recommends the Commission consider modifying the previously approved V2G export opportunities to incorporate this exemption for near-term opportunities, including PG&E's VGI Pilots and PG&E's Commercial EV DAH RTP Export Rate Pilot.

IV. CONCLUSION.

VGIC appreciates the opportunity to submit these reply comments on the Proposed Expansion of the Existing Dynamic Rate Pilots. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

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Date: October 9, 2023